Six Week Review
Fall Prevention
Exoskeleton Clinic

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Our Team

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Project Scope

What are we doing here?

Problem Statement

Slips, trips, and falls are leading causes of injuries, particularly among the elderly and individuals with mobility impairments. Exoskeleton technology has the potential to mitigate these risks by providing real-time support during gait perturbations. Our project focuses on developing a hip and knee exoskeleton capable of detecting and responding to these perturbations, with the goal of improving stability and mobility.

Problem Statement (cont.)

Last semester, preliminary bench and human subject test results demonstrated the the capabilities of the current exoskeleton design. Since then, many areas of improvement have been identified. This semester's goal is to improve upon last year's exoskeleton design and test results with a focus on increasing functionality.

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Deliverables

What have we doing?

Deliverables

Deliverable	Description
Add Hip Flexion Capabilities	— Design hip flexion function — Test cylinder function/range of motion
Thigh Brace Improvements	— Reconfigure strap connections to improve tightness of the brace — Add more straps to add more sturdiness of thigh brace when put all together with hip components.
Abduction Cable	— Redesign abduction cable holder to be more durable and secure to hip brace — Redesign method of connecting cable to moment arm so it can be easily detached and reattached

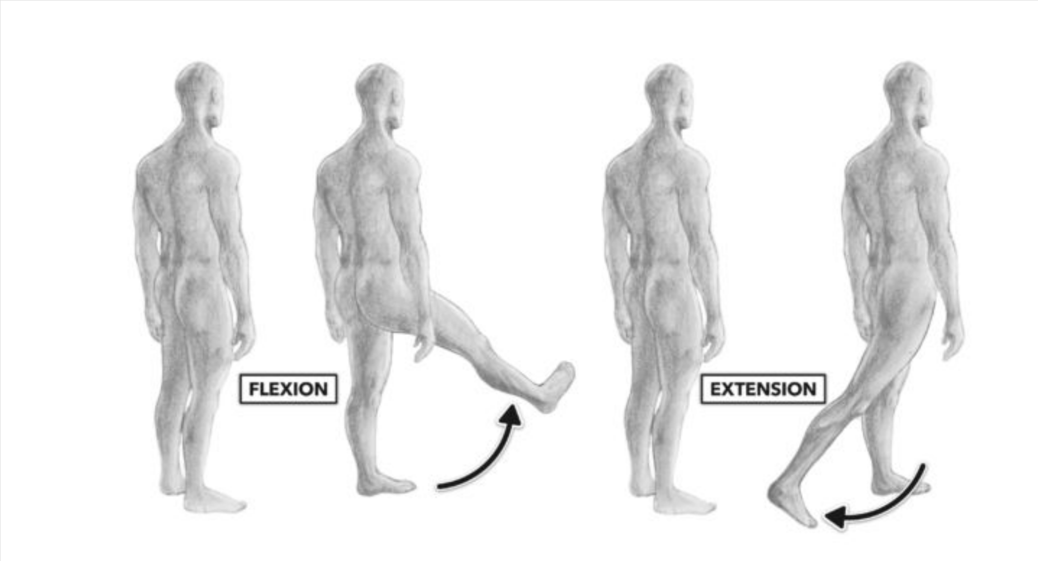
Deliverables

Deliverable	Description
IMU Hardware	— Design of custom IMU harness + PCB for data collection
Control Board PCB	— Custom Control Board PCB for overall system control (solenoids, relays, pressure transducers) and safety features (temperature sensing, current sensing) for emergency shutoff

Hip Flexion Addition

While the leg is currently able to abduct, we are unable to move the leg forward or backward.

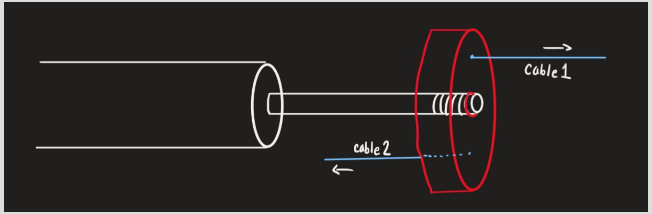
- Helpful for balance recovery and fall prevention
- Provides another degree of freedom for exoskeleton to move

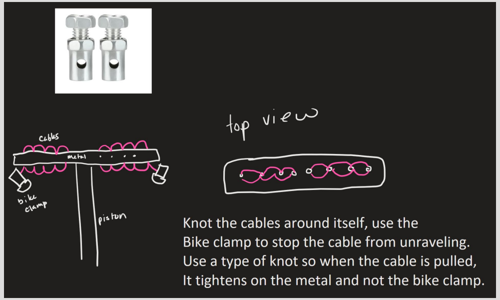


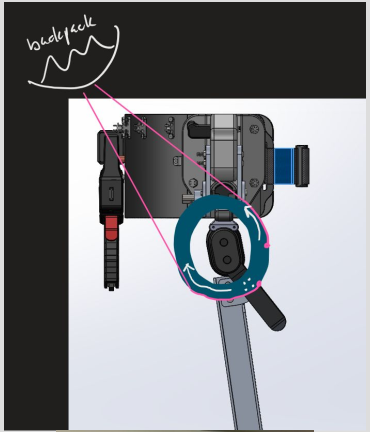
Hip Flexion: Design Ideas and Goals

Preliminary Ideas

- Single piston with a dual cable holder
- Similar movement to knee rotation, using the same mechanism
- Goal: 45-90° motion forward, 15-35° backward
- Ordered 40x75 and 40x100 cylinders to test motion capabilities







Thigh Brace Improvements

While the thigh brace does fit and secures on the thigh, at first it was not tight and would move and loosen up.

- Two extra straps were added to go straight across the brace, while also keeping the x shaped strap configuration.
- When tested, the straps improved the tightness of the brace, and also improved the sturdiness of the brace, while also preventing the straps from loosening from moving

Thigh Brace Design Ideas

Ratchet Mechanism

- Similar to a bike/construction helmet, a simple ratchet tightening mechanism could make putting on the thigh brace easier, as having to clip 8 different straps together can be tedious.

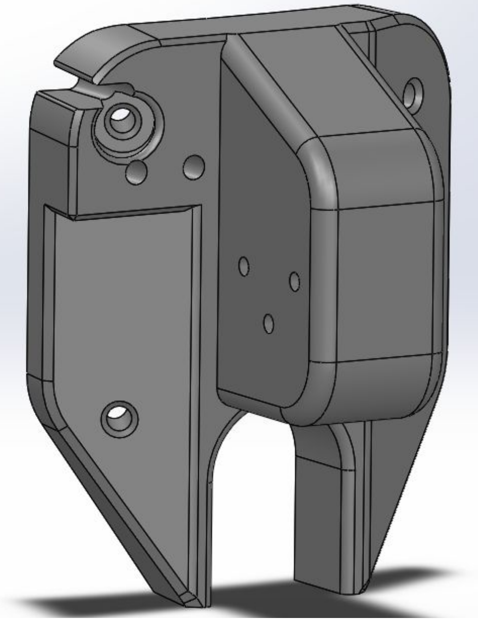
Abduction Cable

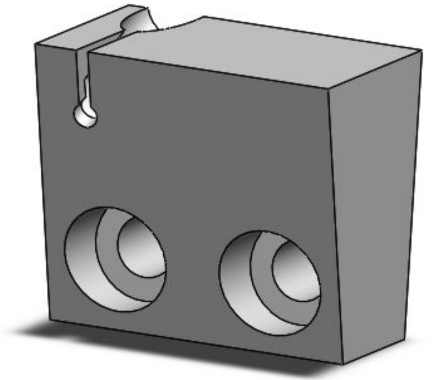
Problems

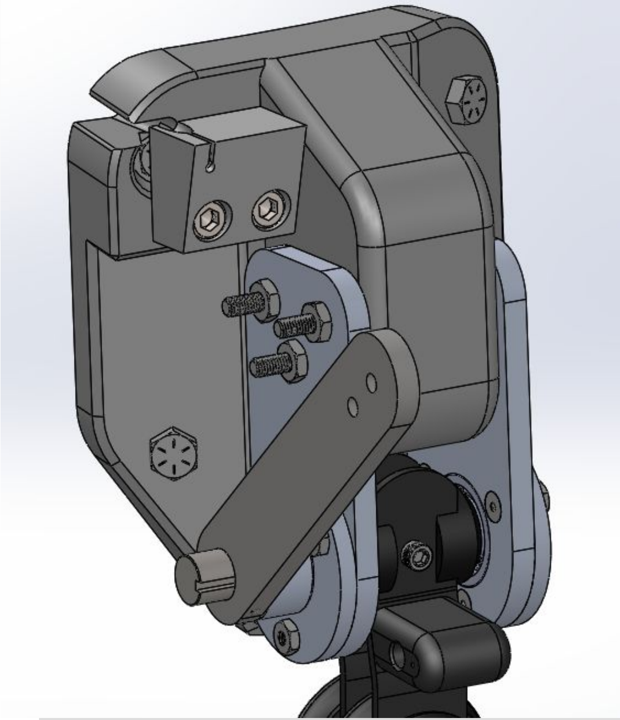
- Steel cable wears down holder
- Sleeve can easily pop out of place

Possible Solutions

- Make cable holder out of steel
- Increase thickness of the front wall
- Add relief hole to prevent wear
- Change material around the hole to TPU
- Add screws to front
- Secure holder with hip brace bolt



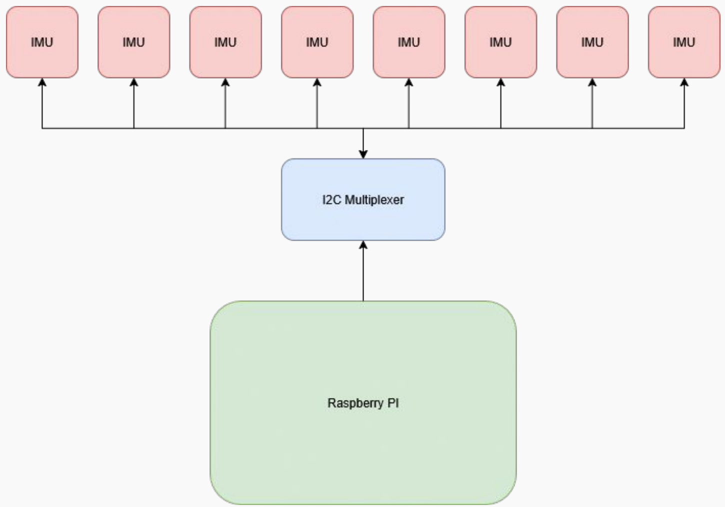


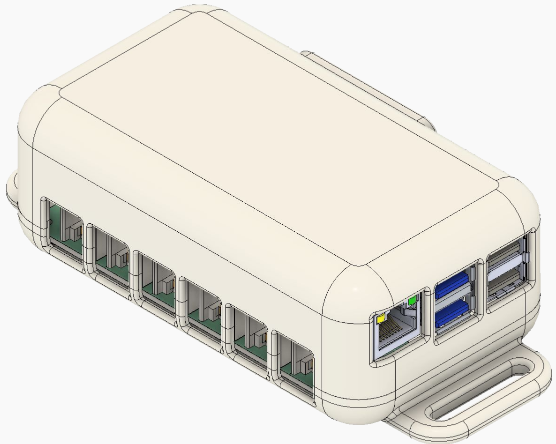


IMU Testing

Key Features:

- Manage 8 IMUs simultaneously via I²C
- Modular RJ12 connectors for each IMU
- Provide appropriate signal filtering and minimize dropout at greater wire lengths
- Interface with a Raspberry Pi 4
- Compact, wearable form factor



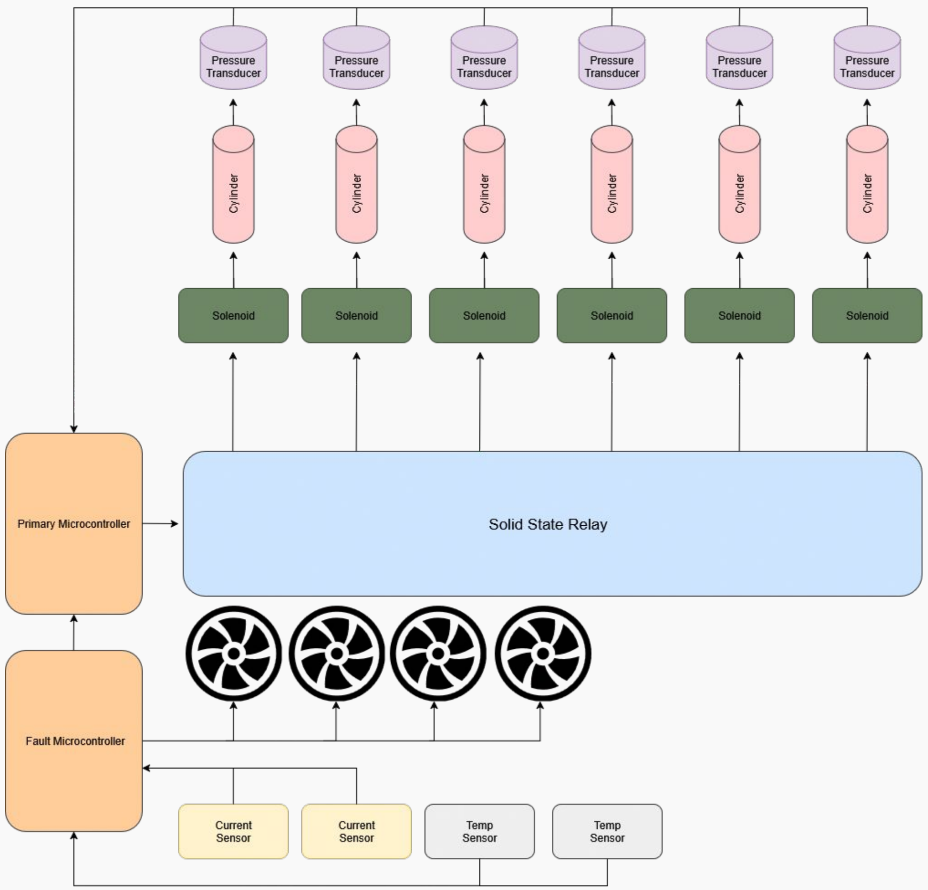


IMU PCB in full mechanical assembly. Strap fastening points on either end for wearability during data acquisition.

Exoskeleton Control System

Key Features:

- Drive solenoids through relay control
- Control cooling fans
- On-board current sensing for input current
- Interface with Adafruit current sensor boards and pressure transducer modules (COTS)
- Distribute power safely and reliably
- Fit within wearable exoskeleton chassis
- Include both a primary microcontroller (basic functions) and a secondary microcontroller (fault detection)



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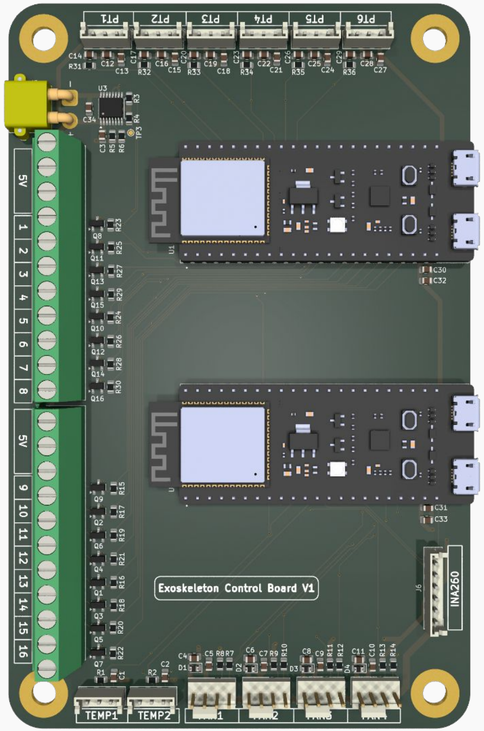

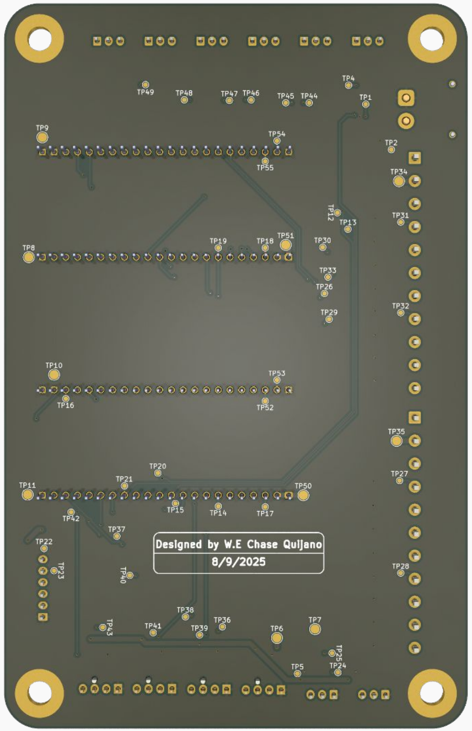

Thank you!

Any Questions?

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Appendix

Standards and Regulations

Standard	Description
IEC 60601-1	Safe power distribution, grounding, and insulation
CISPR 11, EN 61000	Ensure no interference with other devices
ISO/IEC 14908	Data Exchange
ISO 5725/ ISO 6789	Maintain sensor precision/calibration
ISO 14791	Risk management and safety evaluations
ISO 13850	Emergency stop/disconnect
Pilz 8.3.3, 8.5.2	Installation of pneumatic linear actuators

*This is only some of the many standards we follow for this project

Muscle Activation During Slip & Slip Recovery

OpenSim

A free software program that allows for the modeling of mechanical and biological structures

Goal

To use data collected from IMU’s to create a virtual model of the musculoskeletal system, during slips & recovery.

How It Works

Reading Data

The IMU data is read and converted in OpenSim files

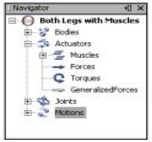
Post Processing

Rotate lab coordinates to match OpenSim coordinate system. Compute force center of pressure

Results

- Perform inverse kinematics
- Perform inverse dynamics
- Use that to calculate which muscles activate and by how much

OpenSim GUI

GUI Components	
Title	**Screenshot**
Toolbar	
Motion Text Box	
Motion Slider / Video Controls	
View Window	
Navigator Window	
Coordinates Window	
Muscle Editor